



BALL STATE
UNIVERSITY

Module 4 Lecture - Discrete Random Variables and Distributions

Introduction to Statistical Methods

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1 Overview and Introduction

1.1 Textbook Learning Objectives

- Recognize and understand discrete probability distribution functions, in general.
- Calculate and interpret expected values.
- Recognize the binomial probability distribution and apply it appropriately.
- Classify discrete word problems by their distributions.

1.2 Instructor Learning Objectives

- Refresh on understanding of behavior and characteristics of discrete variables
- Understand the how to write probability distribution function tables and use them in calculation
- Be able to continue to write coherent and readable notation for probability problems

1.3 Introduction

- We will be expanding on what we spoke about with _____ in the last lecture, but now explicitly applying those ideas to variables

 Discuss: Try describing discrete variables in your own words, based on what you know from Module 1

- We previously described _____ data, or that which is numeric and “count-able”, that has no intervals between integers.
 - Those will be the focus of this lecture, before we focus on _____ variables in the next lecture
- In this lecture, we will introduce the concept of **random variables**, or those that can vary in subsequent _____ (used in the sense as how it was introduced during the probability lecture)
- The _____ of random variables is as follows:

- Uppercase alphabetical character, e.g. X, Y, Z , etc. → written/verbose description of variable
- Lowercase alphabetical equivalent, e.g. x, y, z , etc. → the possible values X can take on
- Example from book:
 - X = the number of heads you get when you toss three _____ coins (concrete description of what the variable is)
 - $x = 0, 1, 2, 3$ (what values it could be)
 - This is a discrete, random variable, because x has _____ different values (random) and because those different values follow the discrete rules, i.e. “countable” in the sense that it is a number of times and no intervals between integers

? Review: What is a synonym for 'fair' when discussing probability?

- A) Equally similar
- B) Similar
- C) Equivalent
- D) Equally likely

Explanation:

2 Probability Distribution Function (PDF) for a Discrete Random Variable

2.1 Introduction

- A **probability distribution function** or PDF is a very scary way to say the list of possible x values and their respective _____
 - The _____ they follow are simple:
 1. All probabilities must sum up to 1, or $\sum P(x) = 1$
 2. Each probability is between 0 and 1, inclusive, or $0 \leq P(x) \leq 1$

 Discuss: Review: try writing the mathematical notation for probability of event A equaling a 30% chance of occurring

- In a probability distribution function, we can think of each _____ value of X , i.e., the x s, having some specified probability of occurring

2.2 PDF Tables

- Example:
 - A survey is being done on how many vehicles families own. Assume the following probability distribution function:
 - X = the number of vehicles owned by a family
 - $x = 0, 1, 2, 3, 4$

x	$P(X = x)$
0	0.10
1	0.20
2	0.30
3	0.25
4	0.15

or...

x	$P(x)$
0	$P(x = 0) = 10/100$
1	$P(x = 1) = 20/100$
2	$P(x = 2) = 30/100$
3	$P(x = 3) = 25/100$
4	$P(x = 4) = 15/100$

- In this example, If I were to randomly select a single family from this distribution, I would have a 30% chance of picking a family with 2 cars

 Discuss: On 30% of days I bring one umbrella with me, and on 70% of days I bring no umbrella with me. Write out a PDF table as above to demonstrate this probability distribution function

- There are 4 types of distributions for discrete random variables:
 - _____ Distributions
 - _____ Distributions
 - _____ Distributions
 - _____ Distributions
 - Each of these has their place, but the last 3 come into play more on more _____ statistical techniques - we'll focus just on the binomial distribution for this lecture

Important

Just because I leave these distribution out in this lecture, doesn't mean that they aren't important! It's mostly just that the others won't be as useful immediately to beginners in statistics.

3 Mean or Expected Value and Standard Deviation

3.1 Introduction

- With PDFs, we sometimes may wish to find the **expected value**, or the “**long-term**” **average or mean**. Thus, doing running this experiment over and over again, we'd expect to _____ on this expected mean
- This is based in the **Law of Large Numbers** a topic alluded to several times in the previous modules
 - Somewhat review from the _____ module: this law states that relative observed frequency approaches the theoretical probability as the number of experiments or trials increases
- For a discrete probability function:

$$\mu = \sum (x \cdot P(x))$$

$$\sigma = \sqrt{\sum [(x - \mu)^2 \cdot P(x)]}$$

- If outcomes for the experiment are equally likely then these formulas work to find the _____ mean and standard deviation for each of the outcomes

 Discuss: Why would we use the mu and sigma notation here, instead of x bar and lowercase s? Do they represent statistics or parameters?

3.2 Mean Calculation Example

Car example from earlier:

x	P(x)	x * P(x)
0	0.10	0.00
1	0.20	0.20
2	0.30	0.60
3	0.25	0.75
4	0.15	0.60

$$\mu = \sum (x * P(x)) = 2.15$$

3.3 Standard Deviation Calculation Example

Car example from earlier:

x	P(x)	x * P(x)	(x - mu)^2 * P(x)
0	0.10	0.00	0.462250
1	0.20	0.20	0.264500

x	$P(x)$	$x * P(x)$	$(x - \mu)^2 * P(x)$
2	0.30	0.60	0.006750
3	0.25	0.75	0.180625
4	0.15	0.60	0.513375

$$\sigma = \sqrt{\sum (x - \mu)^2 * P(x)} = 1.1948$$

3.4 Section Conclusion

- Probability distribution functions serve a purpose - that is - they can describe a particular _____ in the probability of outcomes in the data.
- Once we understand what pattern the variable fits, we can use this information for other _____, as there are the 4 specialty distributions mentioned earlier: geometric, hypergeometric, poisson, and binomial

4 Binomial Distribution

4.1 Introduction

- The binomial distribution has a certain, _____ number of trials represented as n
 - Each trial is _____ and does not affect subsequent trials
- There are two possible _____ in the binomial distribution, “success” and “failure”
 - $P(\text{success}) = p$
 - $P(\text{failure}) = q$
 - $p + q = 1$
- The experiment described above fits the **binomial probability distribution**. Where the random discrete variable X represents the number of successes obtain in n trials
- For the binomial probability distribution:

$$\mu = n * p$$

$$\sigma^2 = n * p * q$$

$$\sigma = \sqrt{n * p * q}$$

4.2 Notation

$$X \sim B(n, p)$$

- This formula is read as “X is a random variable with a binomial distribution”, where n and p represent the same things as before

5 Conclusion

5.1 Recap

- In this shorter lecture, we introduced the concept of discrete random variables, and how they can be represented with probability distribution functions
- We played around with calculating expected mean and standard deviations for outcomes from the probability distribution function table
- We introduced the binomial distribution as a special, specific pattern for a probability distribution represented via “successes” and “failures”

The instructor-provided glossary may not include all terms worth memorizing, make sure you consider using the vocabulary list in your book and your own judgment to make sure you have all relevant terms